



SEMESTER LONG INTERNSHIP – AUTUMN 2026
INTERNSHIP SCREENING TASK REPORT
Task 7 – PCB Design using eSim

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PCB – STM32F4VE

1. Project Overview:

This project presents the complete printed circuit board (PCB) design and schematic files for a custom STM32F4VE microcontroller development board. The design was developed as part of the screening task for the FOSSEE Autumn Internship 2026 program conducted by IIT Bombay.

The primary objective of this project was to design a fully functional, manufacturable development board centered around the STM32F407VET6 microcontroller. The board replicates the core functionality of commercially available STM32 development boards while incorporating a custom 4-layer PCB stackup optimized for signal integrity and power distribution.

Key Project Details:

Category	Specifications & Details
Microcontroller	STM32F407VET6 (ARM Cortex-M4 with FPU, 168 MHz, 512 KB Flash, 192 KB SRAM)
Design Tools	eSim 2.5 (schematic capture) KiCad 6 (PCB layout, integrated via eSim)
Component Libraries	All 3D models, schematic symbols PCB footprints sourced from SnapEDA
Target Manufacturer	JLCPCB (4-layer base model manufacturing rules)
Verification Status	Both ERC and DRC passed with zero errors

The finished design includes complete Gerber files ready for fabrication, a verified 4-layer stackup, and a 3D model for visual verification of component placement and mechanical fit.

2. Circuit Description:

The finished design includes complete Gerber files ready for fabrication, a verified 4-layer stackup, and a 3D model for visual verification of component placement and mechanical fit.

2.1 Power Supply Section

The board operates on a 3.3V power rail, distributed through a dedicated inner power plane (Layer 3). A regulated 3.3V supply is routed to all active components, decoupling capacitors, and peripheral interfaces. A CR12 coin cell backup battery circuit is included to maintain the real-time clock (RTC) and backup SRAM during main power loss. Bulk and bypass capacitors are placed close to the MCU power pins to minimize supply noise and ensure stable operation at high clock frequencies.

2.2 Clock and Timing

The board incorporates onboard crystal oscillators to provide accurate clock references for the microcontroller. A high-speed external (HSE) crystal oscillator provides the primary system clock, while a low-speed external (LSE) crystal is connected for the RTC. Both oscillators include appropriate load capacitors as specified in the STM32F407 datasheet.

2.3 Connectivity and Interfaces

- **Micro USB Port:** Provides a USB 2.0 Full-Speed interface for programming, debugging, and data communication. The USB data lines include appropriate series resistors for impedance matching.
- **SD Card Reader:** Connected via the SDIO peripheral interface, supporting high-speed data transfer to and from SD/microSD cards.
- **Connector Headers:** Extensive pin headers break out the GPIO, I2C, SPI, USART/UART, and CAN bus interfaces, allowing users to connect external sensors, actuators, and communication modules.

2.4 User Interface

The board includes user-accessible push-button switches for reset and user-defined functions, along with onboard status LEDs for power indication and general-purpose output feedback.

2.5 Peripheral Summary

The STM32F407VET6 provides the following on-chip peripherals, all of which are routed and accessible through the board's connectors:

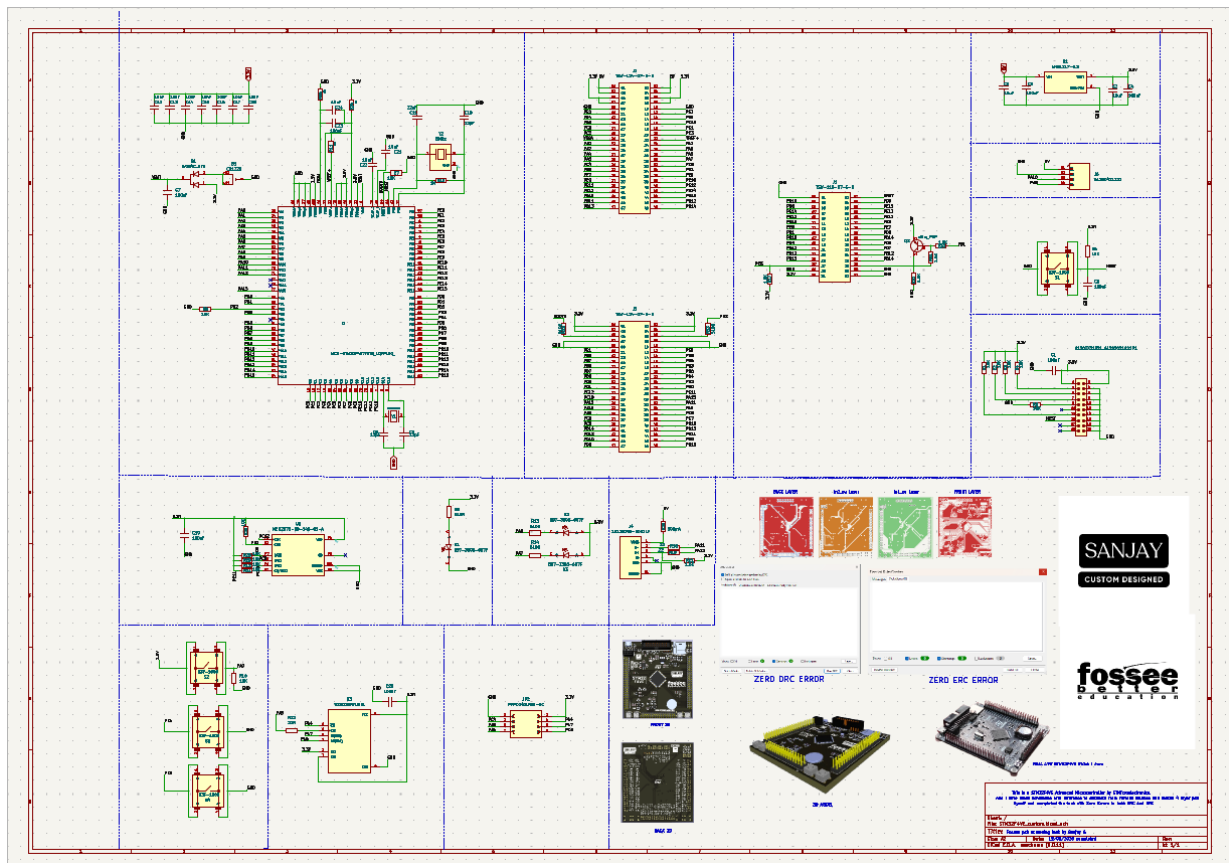
Interface Peripheral	Specification
I2C	Up to 3× buses
SPI	Up to 3× buses
I2S	Up to 2× interfaces
Serial Ports	4× USART + 2× UART
CAN Bus	2× interfaces
SDIO	1× interface
Other Peripherals	Multiple timer channels and ADC/DAC converters

3. Schematic Screenshots

The complete schematic was designed in eSim 2.5 and organized into logical functional blocks for clarity and maintainability.

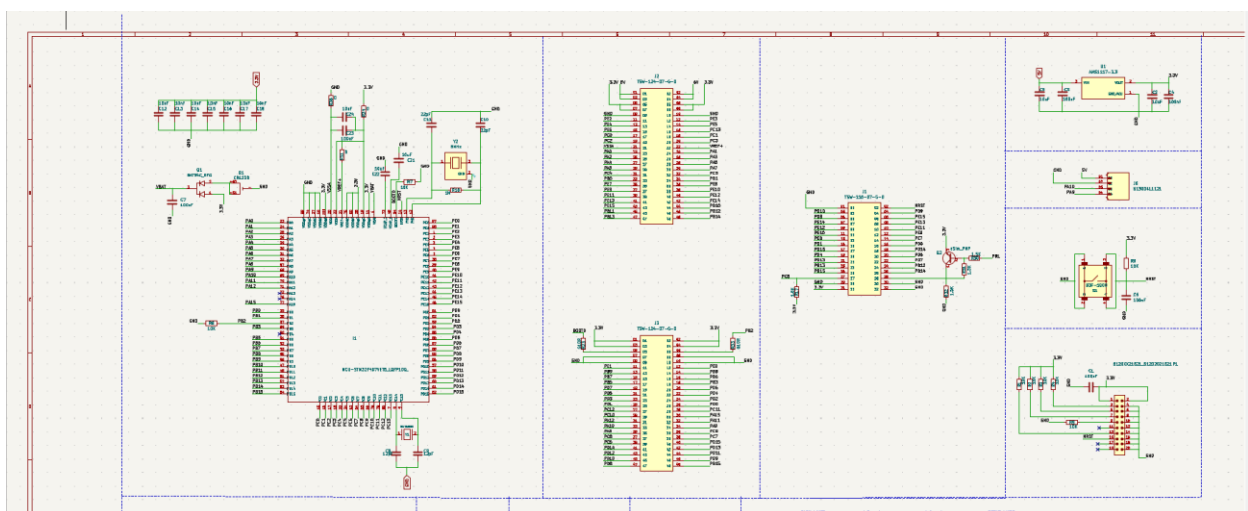
3.1 Full Schematic Overview

The full schematic sheet provides a top-level view of the entire circuit, showing all major blocks — power supply, microcontroller, oscillators, USB, SD card, connectors, and user interface elements — interconnected through labeled nets and power buses.



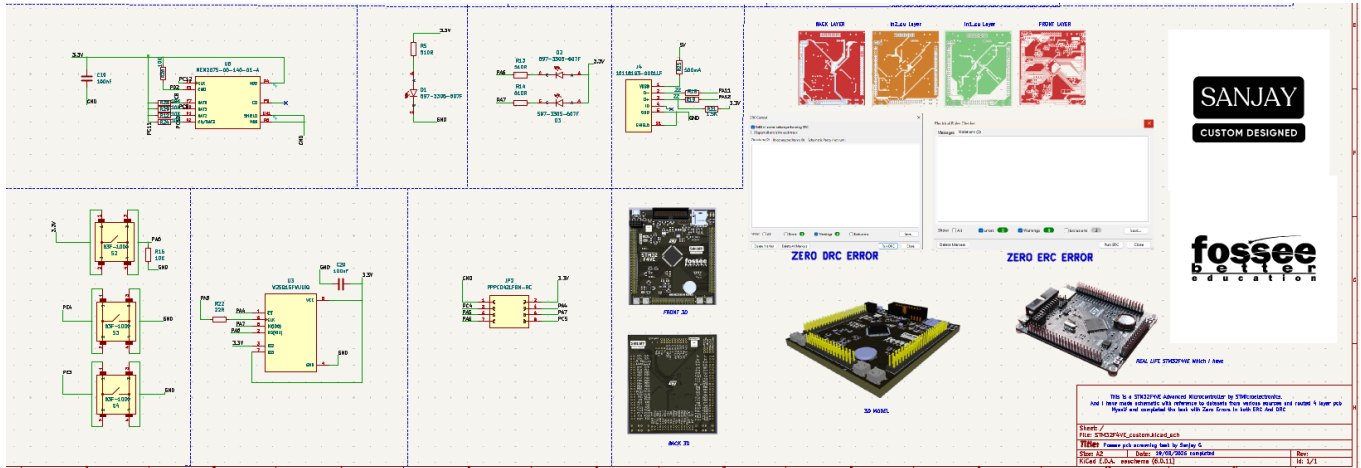
3.2 Schematic Section 1 — MCU and Core Peripherals

This section captures the STM32F407VET6 pin assignments, decoupling capacitor network, crystal oscillator connections (HSE and LSE), reset circuitry, and boot configuration pins. All power pins (VDD, VDDA, VSS, VSSA) are connected to appropriate supply rails with individual bypass capacitors.

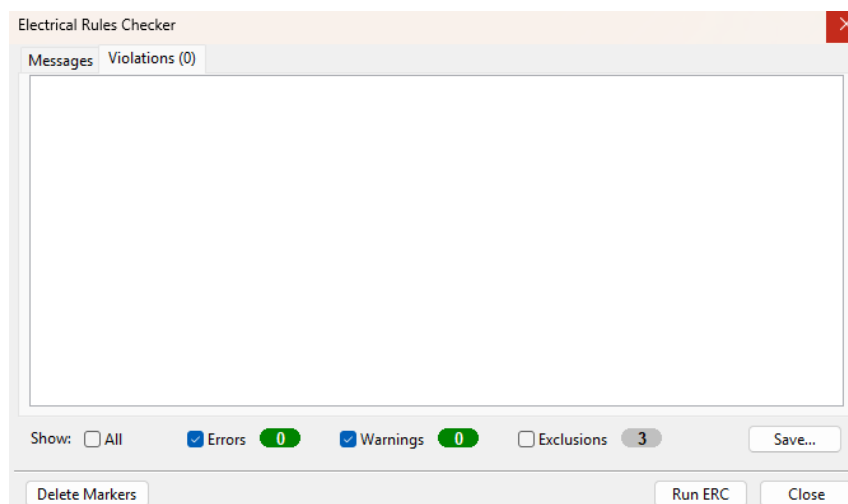
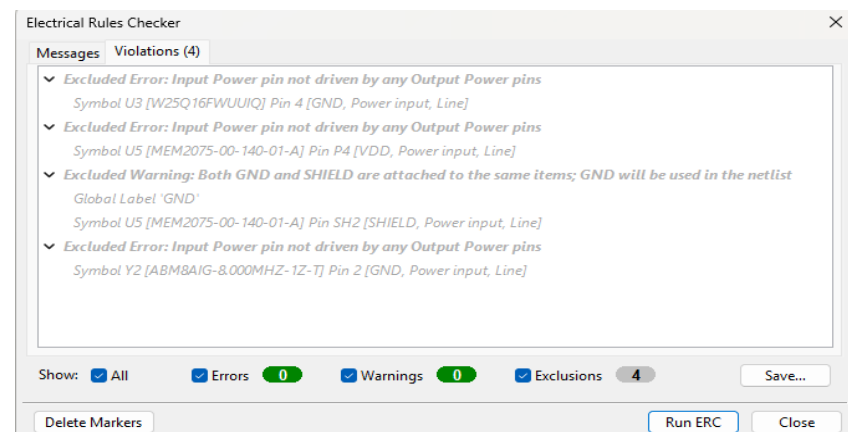


3.3 Schematic Section 2 — Connectors and External Interfaces

This section details the USB interface circuitry, SD card slot connections, GPIO header pinouts, and user interface components (switches and LEDs). The CR12 backup battery circuit and its associated switching network are also shown here.



Note: Wire labels were utilized for power nets throughout the schematic instead of standard KiCad power flags. This approach was chosen for cleaner schematic readability while maintaining correct electrical connectivity. The ERC was configured accordingly to recognize these labeled power nets.



4. Footprint Assignment Details:

All component footprints used in this design were sourced from SnapEDA, ensuring industry-standard pad dimensions, courtyard clearances, and 3D model accuracy.

4.1 Microcontroller

- **Component:** STM32F407VET6
- **Footprint:** LQFP-100 (14×14 mm body, 0.5 mm pin pitch)
- **Source:** SnapEDA verified footprint with 3D STEP model

4.2 Passive Components

- **Resistors:** 0603 (1608 metric) SMD footprint — used for current-limiting resistors, pull-ups/pull-downs, and impedance matching
- **Capacitors:** 0603 (1608 metric) SMD footprint — used for decoupling, bulk bypass, and oscillator load capacitors
- **Crystal Oscillators:** Through-hole and SMD packages as per specific component datasheets

4.3 Connectors

- **Micro USB:** Standard Micro-B USB receptacle footprint with mechanical anchor pads for structural integrity
- **SD Card Slot:** Push-push type micro SD card receptacle with detect pin
- **Pin Headers:** Standard 2.54 mm pitch male pin headers for GPIO and peripheral breakout

4.4 Power Components

- **CR12 Battery Holder:** Through-hole coin cell holder footprint for backup battery
- **Voltage Regulators:** SOT-223 or similar power package footprints with adequate thermal pad area

4.5 User Interface

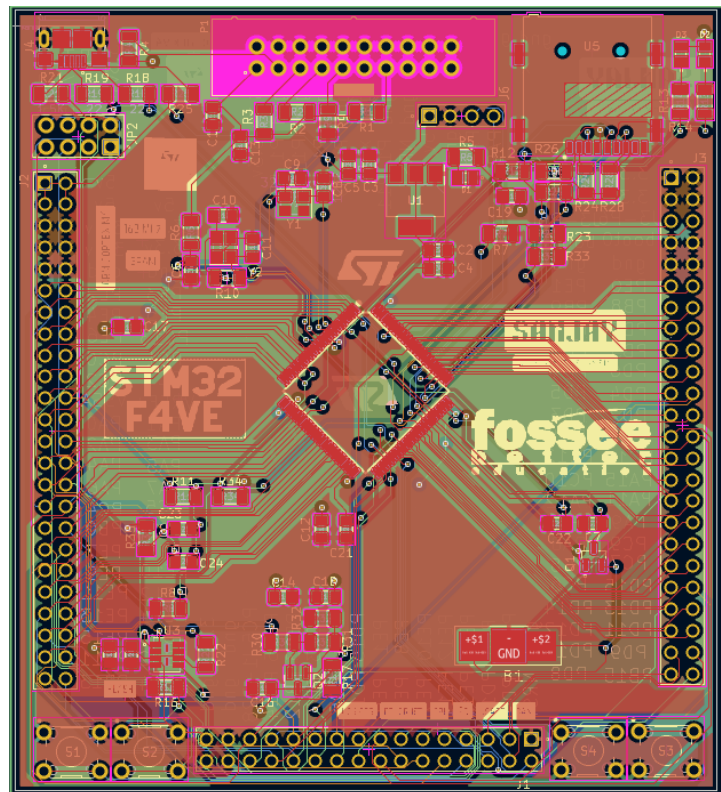
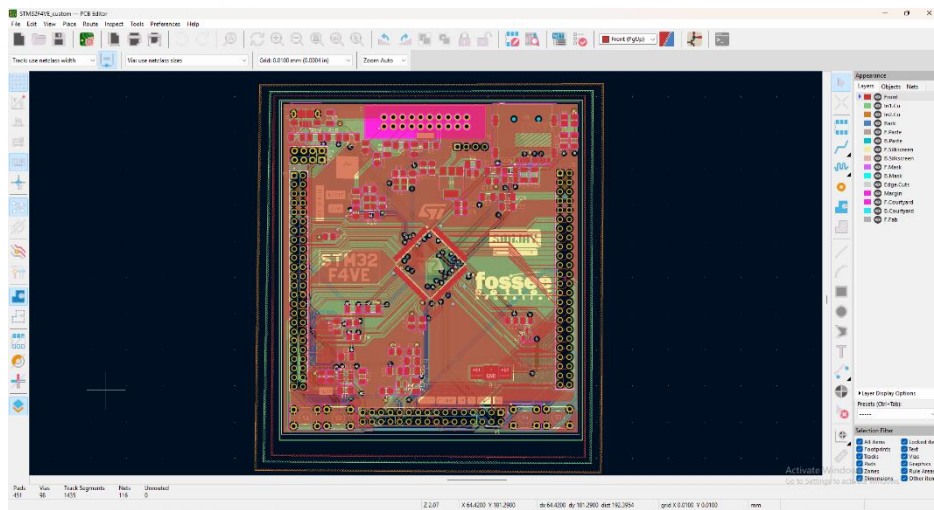
- **Tactile Switches:** Standard 6×6 mm SMD tactile switch footprint
- **LEDs:** 0603 SMD LED footprint

5. PCB Layout:

The PCB layout was completed in KiCad 6 (accessed through eSim integration). The board was designed to comply with JLCPCB 4-layer base model manufacturing constraints.

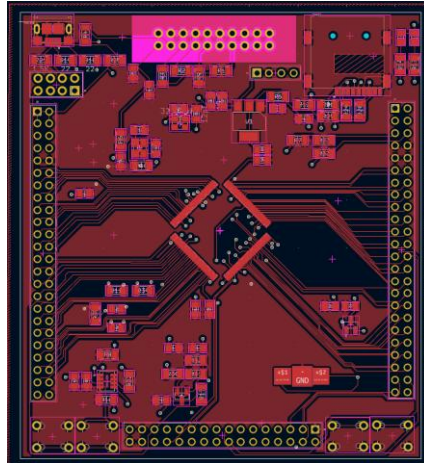
5.1 Complete Board — All Layers View

A composite view showing all copper layers, silkscreen, solder mask, and courtyard outlines overlaid together. This view provides an overview of the overall component placement, routing density, and board outline.



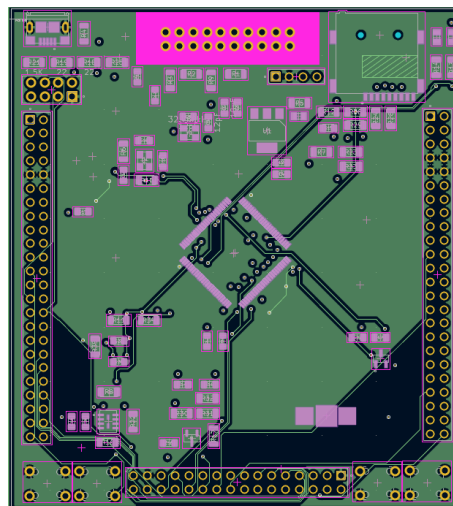
5.2 Front Copper Layer (F.Cu)

The front copper layer carries the primary signal routing between the microcontroller and peripheral components. The ground pour on this layer provides a low-impedance return path for high-frequency signals. Critical traces such as USB data pairs and crystal oscillator connections are routed on this layer with controlled lengths.



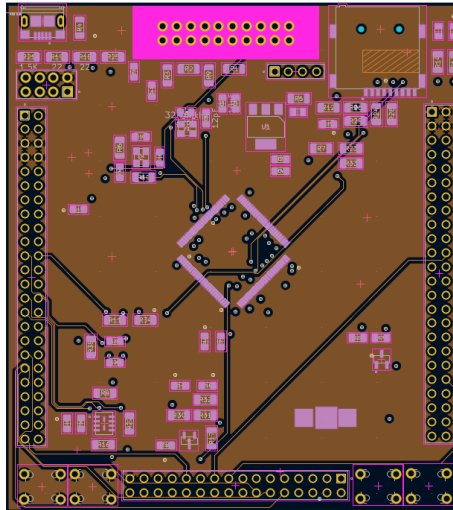
5.3 Inner Layer 1 (In1.Cu) — Ground Plane

A continuous, unbroken ground plane provides a solid reference for signal integrity, reduces electromagnetic interference (EMI), and creates a low-impedance return path for all signal traces on adjacent layers.



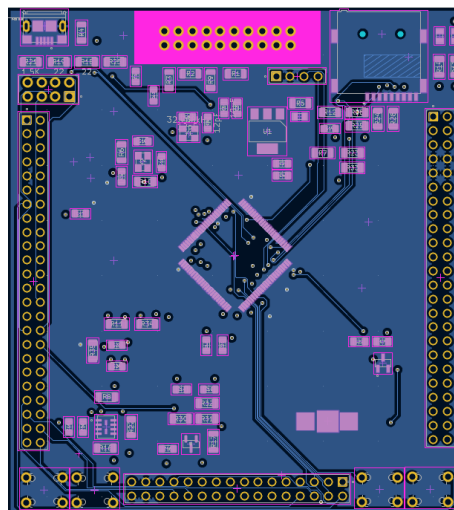
5.4 Inner Layer 2 (In2.Cu) — Power Plane (3.3V)

A dedicated 3.3V power plane distributes clean, low-impedance power to all components. The unbroken plane, paired with the adjacent ground plane, forms a distributed capacitance that helps suppress power supply noise.

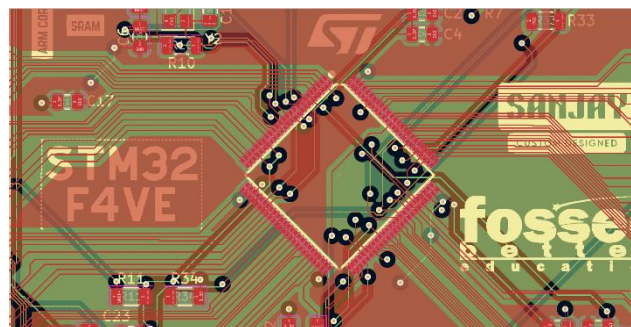
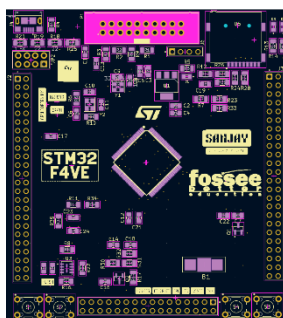


5.5 Back Copper Layer (B.Cu)

The back copper layer serves as an additional ground plane with selective signal routing where necessary. Components placed on the bottom side of the board are connected through this layer.



5.6 Component Placement Views



6. Routing Methodology:

6.1 Layer Stackup Strategy

The board employs a 4-layer stackup specifically designed for mixed-signal applications:

Layer	Name	Function	Purpose
1	F.Cu (Front)	Signal+ Ground Pour	Primary signal routing with ground fill
2	In1.Cu (Inner 1)	Dedicated Ground Plane	Continuous GND reference for signal integrity
3	In2.Cu (Inner 2)	Dedicated Power Plane	Unbroken 3.3V distribution
4	B.Cu (Back)	Ground Plane	Secondary ground, minimal signal routing

This stackup ensures that every signal trace on Layer 1 has an immediate ground reference (Layer 2) directly beneath it, minimizing loop area and improving EMI performance. The power plane on Layer 3, sandwiched between two ground planes, benefits from inherent distributed decoupling capacitance.

6.2 Signal Routing Guidelines

- **Decoupling Capacitors:** Placed as close as physically possible to the MCU power pins, with vias connecting directly to the power and ground planes.
- **Crystal Oscillator Traces:** Routed with short, direct traces to minimize parasitic capacitance and inductance. A ground guard ring surrounds the oscillator area to reduce noise coupling.
- **USB Data Lines:** Routed as differential pairs with matched lengths and controlled impedance. Series resistors placed close to the MCU pins.
- **General Signal Traces:** Routed on the front copper layer wherever possible, utilizing the solid ground plane beneath for a clean return path.
- **Power Routing:** No signal traces cross splits or gaps in the power/ground planes to prevent return path discontinuities.

6.3 Design Rule Constraints

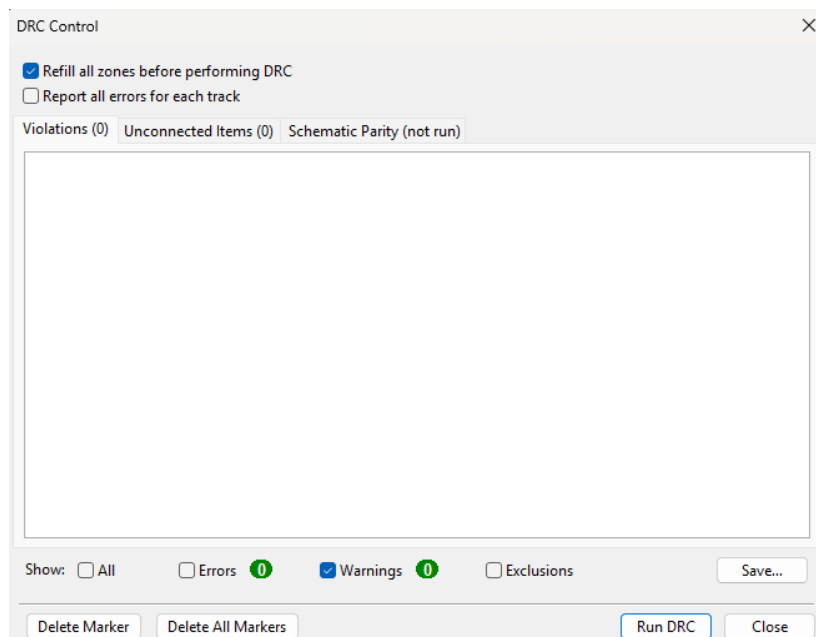
All routing adheres to JLCPCB 4-layer base model specifications:

- Minimum trace width and spacing as per manufacturer capabilities
- Minimum via drill size and annular ring as per JLCPCB tolerances
- Copper-to-edge clearance maintained for board outline routing
- No acid traps or acute-angle traces in the routing

7. DRC Results:

7.2 Design Rules Check (DRC)

The Design Rules Check was performed on the completed PCB layout to verify physical manufacturability and compliance with JLCPCB design rules.

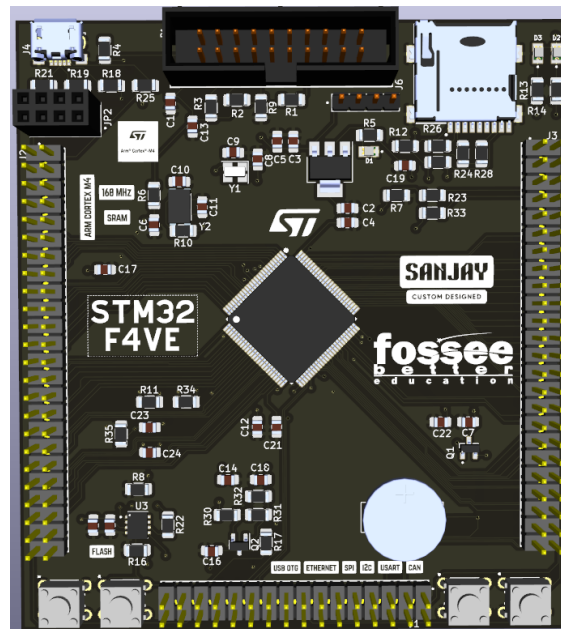


8. 3D Model:

The 3D model of the board was generated using KiCad's built-in 3D viewer with STEP models imported from SnapEDA for all components. The 3D visualization serves to verify component placement accuracy, mechanical clearances, connector accessibility, and overall board aesthetics before committing to fabrication.

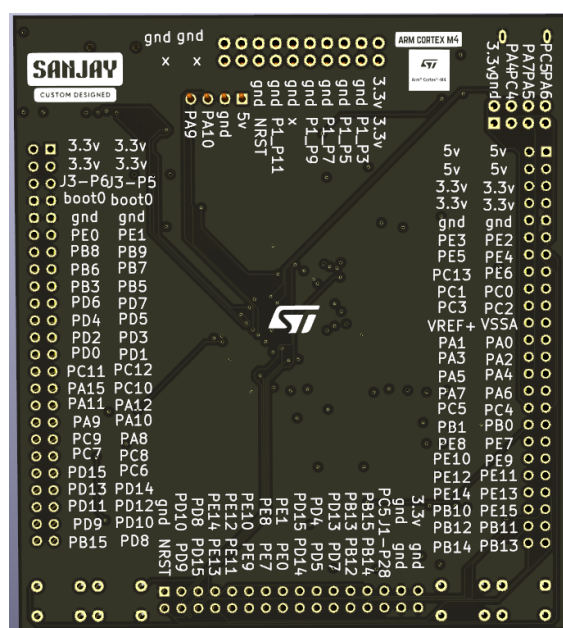
8.1 Front View

The front (top) view shows the primary component side of the board, featuring the STM32F407VET6 in its LQFP-100 package at the center, surrounded by decoupling capacitors, crystal oscillators, connectors, and GPIO headers. The custom silkscreen graphics — including the FOSSEE logo, STM logo, and designer name (Sanjay) — are visible on the front solder mask layer.



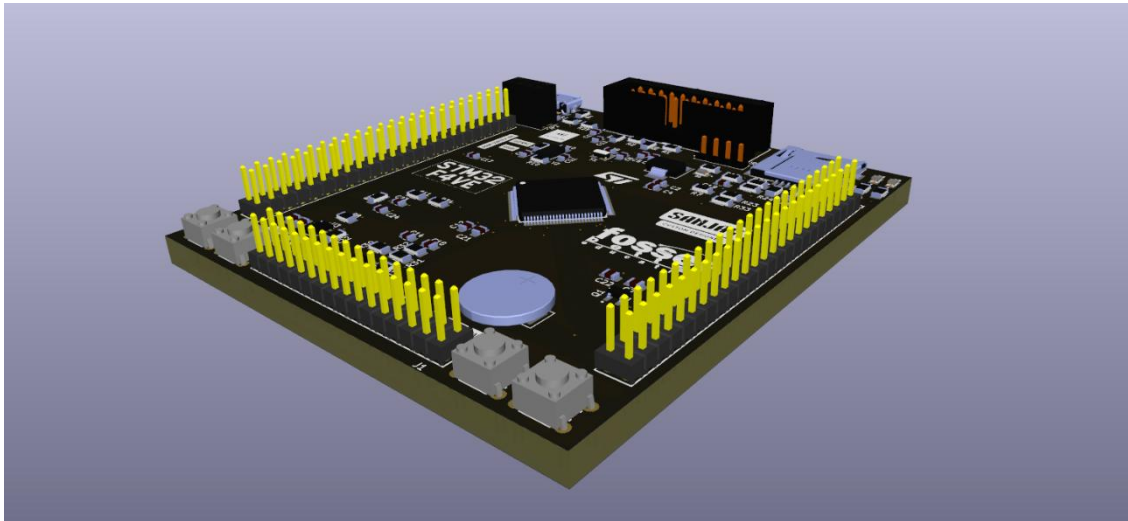
8.2 Back View

The back (bottom) view reveals the underside of the board, showing any bottom-mounted components, the back copper ground plane fill, and additional custom silkscreen artwork applied to the rear of the board.



8.3 Side Views

Side-angle views provide a perspective on the board's vertical profile, showcasing connector heights, component clearances, and the overall mechanical form factor. These views are critical for verifying that tall components (USB connectors, pin headers, SD card slot) do not interfere with enclosure constraints.



Design Note: The board features a **black solder mask** finish, chosen to replicate the appearance of standard commercial STM32 development boards. The black mask also improves silkscreen contrast and overall visual aesthetics.

9. Design Consideration and Conclusion:

9.1 Design Considerations

Power Integrity: The dedicated inner power and ground planes (Layers 2 and 3) provide low-impedance power distribution and minimize supply noise. Decoupling capacitors at every MCU power pin, combined with the inherent plane-to-plane capacitance of the stackup, ensure stable power delivery at the 168 MHz operating frequency.

Signal Integrity: The 4-layer stackup ensures every signal trace on the front copper layer has an uninterrupted ground reference directly beneath it. This minimizes signal return loop area, reduces crosstalk between adjacent traces, and improves overall electromagnetic compatibility (EMC). Critical signals (USB, clocks) were routed with careful length matching and impedance control.

Thermal Management: The ground and power planes serve as effective heat spreaders, distributing thermal energy from the MCU and voltage regulators across

a large copper area. The exposed pad (if applicable) of the MCU connects directly to the ground plane through multiple thermal vias.

Manufacturability: All design rules strictly adhere to JLCPCB 4-layer base model specifications, ensuring the board can be fabricated without custom manufacturing tolerances or premium process requirements. The board passes DRC with zero errors, confirming that every feature — traces, vias, clearances, and drill holes — is within the manufacturer's standard capabilities.

Component Sourcing: By using SnapEDA for all component libraries, the design benefits from verified, datasheet-accurate footprints and 3D models. This reduces the risk of footprint errors that commonly cause fabrication or assembly failures.

Aesthetics and Branding: Custom silkscreen graphics featuring the FOSSEE logo, STM logo, and designer's name add a professional finish to the board. The black solder mask choice aligns with the visual identity of commercial STM32 products.

9.2 Conclusions

This project successfully delivers a complete, manufacturable PCB design for an STM32F407VET6 development board. The design achieves the following key outcomes:

1. **Fully verified design** — Both ERC and DRC passed with zero errors, confirming electrical correctness and physical manufacturability.
2. **Professional-grade 4-layer stackup** — Dedicated power and ground planes ensure robust power integrity and signal quality, meeting the demands of a 168 MHz ARM Cortex-M4 system.
3. **Comprehensive connectivity** — The board exposes all major MCU peripherals (USB, SDIO, I2C, SPI, USART, CAN) through accessible connectors and headers, making it a versatile development platform.
4. **Production-ready output** — Gerber files are generated and verified against the JLCPCB online viewer, ready for immediate fabrication.
5. **Complete documentation** — Schematics, PCB layouts, layer renders, 3D models, and verification reports are all included in the repository for full design transparency.

The design demonstrates proficiency in schematic capture, multi-layer PCB layout, design rule compliance, and 3D visualization — the core competencies required for the FOSSEE Autumn Internship 2026 screening task.

